NATIONAL CANCER INSTITUTE

Bladder Cancer Treatment

Different types of treatment are available for bladder cancer. You and your cancer care team will work together to decide your treatment plan, which may include more than one type of treatment. Many factors will be considered, such as the stage and grade of the cancer, your overall health, and your preferences. Your plan will include information about your cancer, the goals of treatment, your treatment options and the possible side effects, and the expected length of treatment.

It will be helpful to talk with your cancer care team before treatment begins about what to expect. Some things you'll want to learn about include what you need to do before treatment begins, how you'll feel while going through it, and what kind of help you will need. To learn more, visit Questions to Ask Your Doctor About Your Treatment. For treatment by stage, visit Treatment of Bladder Cancer by Stage.

Surgery

Surgery is the main treatment for bladder cancer. The type of surgery depends on where the cancer is located. Other treatments may be given in addition to surgery:

- Treatment given before surgery is called preoperative therapy or neoadjuvant therapy. Chemotherapy may be given before surgery to shrink the tumor and reduce the amount of tissue that needs to be removed during surgery.
- Treatment given after surgery, to lower the risk that the cancer will come back, is called adjuvant therapy. After the doctor removes all the cancer that can be seen, some patients may be given chemotherapy, radiation therapy, immunotherapy, and/or targeted therapy to kill any cancer cells that are left.

Learn more about Surgery to Treat Cancer.

The types of surgery done to treat bladder cancer are:

Transurethral resection (TUR) with fulguration

During TUR with fulguration, the doctor inserts a cystoscope (a thin lighted tube) into the bladder through the urethra. A tool with a small wire loop on the end is then used to remove the cancer or to burn the tumor away with high-energy electricity. This is known as fulguration.

Partial cystectomy

Partial cystectomy is surgery to remove part of the bladder. This may be done for patients who have a low-grade tumor that has invaded the wall of the bladder but is limited to one area of the bladder. Because only a part of the bladder is removed, patients are able to urinate normally after recovering from this surgery. This is also called segmental cystectomy.

Radical cystectomy with urinary diversion

Radical cystectomy is surgery to remove the bladder and any lymph nodes and nearby organs that contain cancer. This surgery may be done when the bladder cancer invades the muscle layers or when non-muscle-invasive bladder cancer involves a large part of the bladder:

- In men, the nearby organs that are removed are the prostate and the seminal vesicles.
- In women, the uterus, the ovaries, and part of the vagina are removed.

Sometimes, when the cancer has spread outside the bladder and can't be completely removed, surgery to remove only the bladder may be done to reduce urinary symptoms caused by the cancer.

When the bladder must be removed, the surgeon performs a procedure called urinary diversion to create another way for the body to store and pass urine. It may involve redirecting urine into the colon, using catheters to drain the bladder, or making an opening in the abdomen that connects to a bag outside the body for collecting urine. To learn more, visit the National Institute of Diabetes and Digestive and Kidney Diseases page on Urinary Diversion.

Radiation therapy

Radiation therapy uses high-energy x-rays or other types of radiation to kill cancer cells or keep them from growing. Bladder cancer is sometimes treated with external beam radiation therapy. This type of radiation therapy uses a machine outside the body to send radiation toward the area of the body with cancer. Radiation therapy may be given alone or with other types of treatment, such as chemotherapy.

Learn more about External Beam Radiation Therapy for Cancer and Radiation Therapy Side Effects.

Chemotherapy

Chemotherapy (also called chemo) uses drugs to stop the growth of cancer cells, either by killing the cells or by stopping them from dividing. Chemotherapy may be given alone or with other types of treatment. The way the chemotherapy is given depends on the type and stage of the cancer being treated.

Systemic chemotherapy

Systemic chemotherapy for bladder cancer is when chemotherapy drugs are injected into a vein. When given this way, the drugs enter the bloodstream to reach cancer cells throughout the body. Systemic chemotherapy drugs that may be used to treat bladder cancer are:

- carboplatin
- cisplatin
- doxorubicin
- fluorouracil (5-FU)
- gemcitabine
- methotrexate
- mitomycin
- paclitaxel
- vinblastine

Combinations of these drugs may be used. Other chemotherapy drugs not listed here may also be used.

Intravesical chemotherapy

For bladder cancer, chemotherapy may be intravesical, meaning it is put into the bladder through a tube inserted into the urethra. Intravesical treatments flush the bladder with drugs that kill cancer cells that remain after surgery. This lowers the chance of the cancer coming back.

Mitomycin and gemcitabine are two chemotherapy drugs given as intravesical chemotherapy to treat bladder cancer. These drugs can also be given systemically.

To learn more about how chemotherapy works, how it is given, common side effects, and more, visit Chemotherapy to Treat Cancer and Chemotherapy and You: Support for People With Cancer.

Immunotherapy

Immunotherapy helps a person's immune system fight cancer. Your doctor may suggest biomarker tests to help predict your response to certain immunotherapy drugs. Learn more about Biomarker Testing for Cancer Treatment.

Systemic immunotherapy

Systemic immunotherapy drugs used to treat urothelial cancer (a type of bladder cancer) include:

- atezolizumab
- avelumab
- nivolumab
- pembrolizumab

These drugs work in more than one way to kill cancer cells. They are also considered targeted therapy because they target specific changes or substances in cancer cells (visit the section on Targeted therapy).

Intravesical immunotherapy

BCG (bacillus Calmette-Guérin), nadofaragene firadenovec-vncg, and nogapendekin alfa inbakicept-pmln are intravesical immunotherapy drugs used to treat bladder cancer. They are given in a solution that is placed directly into the bladder using a catheter (thin tube). Intravesical treatments flush the bladder with drugs that kill cancer cells that remain after surgery. This lowers the chance of the cancer coming back.

Immunotherapy uses the body's immune system to fight cancer. This animation explains one type of immunotherapy called nonspecific immune stimulation that is used to treat cancer.

Learn more about Immunotherapy to Treat Cancer and Immunotherapy Side Effects.

Targeted therapy

Targeted therapy uses drugs or other substances to block the action of specific enzymes, proteins, or other molecules involved in the growth and spread of cancer cells. Your doctor may suggest biomarker tests to help predict your response to certain targeted therapy drugs. Learn more about Biomarker Testing for Cancer.

Targeted therapies used to treat bladder cancer include:

- enfortumab vedotin
- erdafitinib
- ramucirumab
- sacituzumab govitecan-hziy

Learn more about Targeted Therapy to Treat Cancer.

Clinical trials

For some people, joining a clinical trial may be an option. There are different types of clinical trials for people with cancer. For example, a treatment trial tests new treatments or new ways of using current treatments. Supportive care and palliative care trials look at ways to improve quality of life, especially for those who have side effects from cancer and its treatment.

You can use the clinical trial search to find NCI-supported cancer clinical trials accepting participants. The search allows you to filter trials based on the type of cancer, your age, and where the trials are being done. Clinical trials supported by other organizations can be found on the ClinicalTrials.gov website.

Learn more about clinical trials, including how to find and join one, at Clinical Trials Information for Patients and Caregivers.

Follow-up care

Some of the tests that were done to diagnose or stage the cancer may be repeated. Some tests will be repeated to see how well the treatment is working. Decisions about whether to continue, change, or stop treatment may be based on the results of these tests. These tests are sometimes called follow-up tests or check-ups.

Updated: September 12, 2024
